Cart lab: change mass

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a controlled comparison to describe the effect of system mass on acceleration.

SUCCESS CRITERIA - I can...

1. Identify total moving mass and keep driving force fixed.
2. Compare acceleration values using a fair test.
3. Separate model data from measured evidence.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A car starts at +24 m/s and brakes at -4 m/s² until rest. A student extends that acceleration to 8 s. Find the stop time/distance and explain why extension is invalid for ordinary braking.

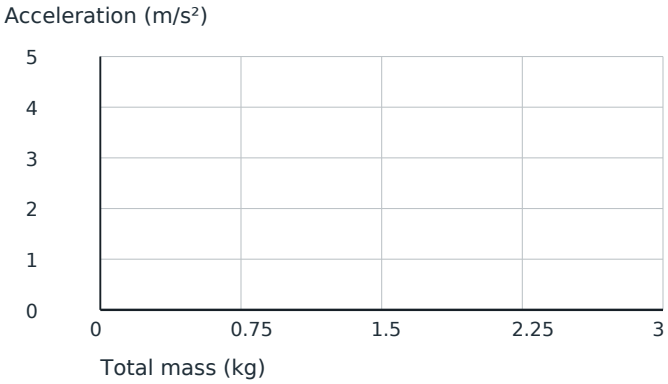
Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $a = F_{net}/M = m_h\,g/M$.
Ideal cart + hanger: M is all moving mass; m_h is hanging mass. Hold m_h fixed; neglect resistance and pulley inertia.

Worked reasoning: Supplied cart trials use the same net force: masses 1 and 2 kg accelerate at 4 and 2 m/s². In both cases ma=4 N; doubling mass halves acceleration.

Your turn: Plot acceleration against mass for the two trials. Label what was held constant.



Marzano check - circle: ① ② ③

03 Test an inverse relationship

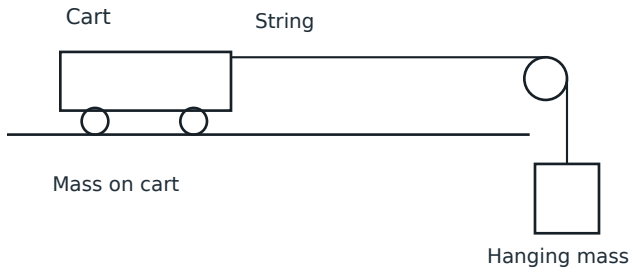
The table contains ideal data for a constant 0.49 N net force. Calculate 1/M for each mass. Sketch acceleration versus 1/M, then state the slope's units and physical meaning.

M (kg)	a (m/s²)
0.50	0.980
0.75	0.653
1.00	0.490
1.50	0.327

Marzano check - circle: ① ② ③

04 System boundary

The cart and hanger accelerate together. Use the setup model to explain why both belong in the total moving mass and why string tension is internal to the combined system. State two idealizations.



Marzano check - circle: ① ② ③

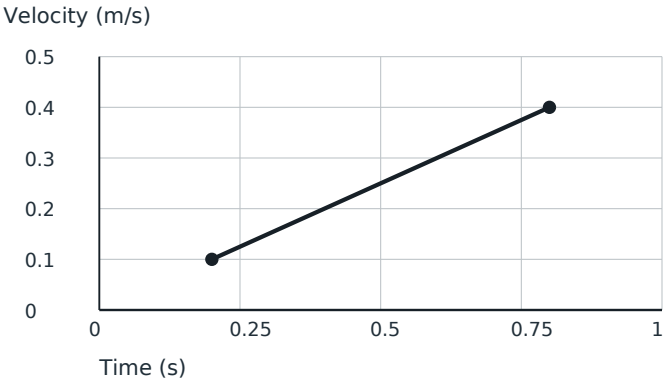
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Continue down each column.

05 Measured slope and repetitions

A sensor records velocity changing from 0.10 m/s at 0.20 s to 0.40 m/s at 0.80 s. Find the acceleration from the graph. Then design a table for three total masses and three trials per mass; label any data you actually measure.



Marzano check - circle: ① ② ③

06 Review: connect an earlier idea

A 3 kg object hangs still from a vertical string. Use g=9.8 N/kg. Name all forces on it with sources, sizes and directions. Does object-on-string belong on this diagram? Draw the forces on the chosen object and label their sources.

Add and label force arrows.

Marzano check - circle: ① ② ③

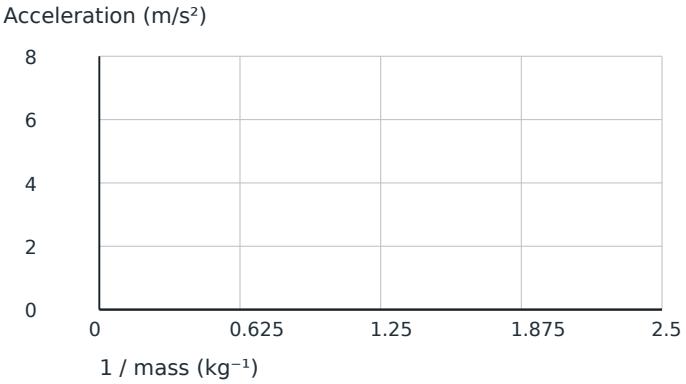
07 Exit check

A 0.80 kg moving system accelerates at 0.60 m/s². At the same net force, predict a for 1.20 kg. Explain how added resistance would change the prediction.

Marzano check - circle: ① ② ③

08 STRETCH: test the model

A cart model predicts a=F/m. At constant net force 3 N, find accelerations for masses 0.5, 1, 2 kg. Choose a transformed horizontal axis that makes a straight-line graph and state its slope.



Marzano check - circle: ① ② ③